◈ **RetailIQ**

RETAIL INTELLIGENCE PLATFORM

NAVIGATION

- Executive Overview
- Demand Forecasting
- Historical vs Forecast
- Model Performance
- Store & Category Analysis
- Inventory Optimization
- High-Risk Products

FILTER DATA

Product Category

HOUSEH... × FOODS × HOBBIES ×

Store

CA_1 × CA_2 × CA_3 × WI_1 × WI_2 × WI_3 ×

Search Product

Search product ID...

RETAIL DEMAND & INVENTORY INTELLIGENCE

Smarter Demand. Better Inventory Decisions.

A data-driven view of retail demand, model performance, product trends and inventory risk.

Displaying 30,490 product-store combinations across 10 stores.

Inventory Optimization

Review demand-based safety stock and reorder thresholds for product-store combinations.

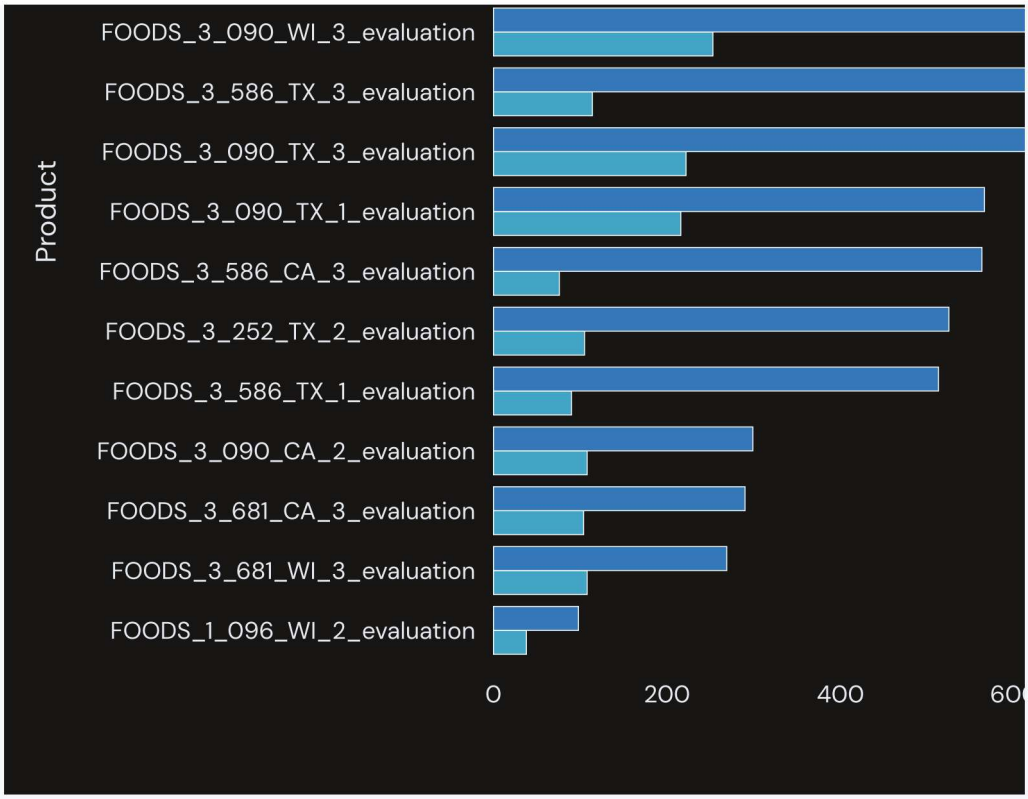
Illustrative planning assumptions: 7-day lead time and 95% service level. These calculations are threshold recommendations, not live inventory positions or purchase orders.

Products Analyzed	Average Safety Stock	Average Reorder Point
30,490	6.79	14.39

Inventory Threshold Comparison



TX_1 ×
TX_2 ×
TX_3 ×



Inventory Recommendations

id	category	store_id	avg_
FOODS_3_090_CA_3_evaluation	FOODS	CA_3	
FOODS_3_586_TX_2_evaluation	FOODS	TX_2	
FOODS_3_090_CA_1_evaluation	FOODS	CA_1	
FOODS_3_090_TX_2_evaluation	FOODS	TX_2	
FOODS_3_090_WI_3_evaluation	FOODS	WI_3	
FOODS_3_586_TX_3_evaluation	FOODS	TX_3	
FOODS_3_090_TX_3_evaluation	FOODS	TX_3	
FOODS_3_090_TX_1_evaluation	FOODS	TX_1	
FOODS_3_586_CA_3_evaluation	FOODS	CA_3	
FOODS_3_252_TX_2_evaluation	FOODS	TX_2	

Download CSV

Methodology and limitations

Safety stock and reorder points are calculated from historical demand variability and assumed replenishment lead time. The M5 dataset does not provide actual on-hand inventory or supplier lead times. Therefore, these outputs should be interpreted as illustrative planning thresholds.

RetailIQ

Retail Demand Forecasting & Inventory Intelligence

Built with Python · Pandas · LightGBM · PyTorch · Streamlit · Plotly

Historical M5 benchmark | May 2016 – June 2016